NETWORKING

WebSockets Visual Guide

Full-duplex over HTTP's bones — real-time messaging for the web.

ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/5

The Upgrade Handshake

Starts as HTTP, then both sides agree to stop speaking HTTP.

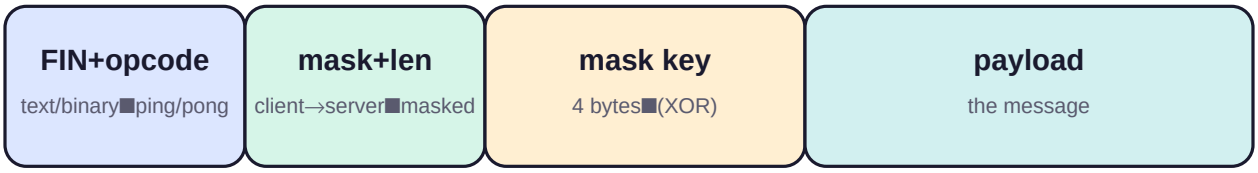
```
GET /chat HTTP/1.1
Upgrade: websocket      Connection: Upgrade
Sec-WebSocket-Key: dGhlIHNhbXBsZSBub25jZQ==

HTTP/1.1 101 Switching Protocols
Sec-WebSocket-Accept: s3pPLMBiTxaQ9kYGzzhZRbK+xOo=
```

- The handshake is plain HTTP — so it passes through existing proxies and firewalls.
- 101 Switching Protocols: from here the TCP connection speaks the WebSocket framing protocol.
- The accept key proves the server understands WebSocket (SHA-1 of key + magic GUID).

STEP 2/5

Frames: The Wire Format

Lightweight envelopes — ~2 bytes overhead per message.

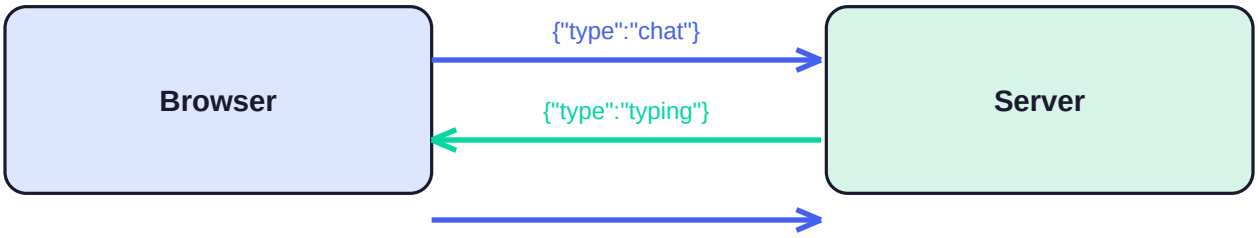


- Opcodes: text (0x1), binary (0x2), close (0x8), ping (0x9), pong (0xA).
- Client frames are MASKED with a random key — defeats cache-poisoning attacks on proxies.
- Ping/pong keep idle connections alive through NATs and load balancers.

STEP 3/5

Full-Duplex Messaging

Either side sends anytime — no request/response turn-taking.



- Chat, live dashboards, multiplayer games, collaborative editing — anywhere latency matters.
- Message-based, not stream-based: each frame is a discrete message your app handles.
- One persistent connection replaces constant polling — far less header overhead.

STEP 4/5

WebSocket vs SSE vs Long-Polling

Three ways to push — pick by direction and complexity.

Long-polling

Client holds a request open; server answers when data arrives. Simple, works everywhere, chatty.

SSE

Server→client text stream over HTTP. Auto-reconnect, simple — but one-directional.

WebSocket

True bidirectional binary/text messaging. Most capable, most connection state to manage.

Decision: need client→server pushes? WebSocket. Server→client feed only? SSE is simpler.

STEP 5/5

Scaling WebSockets

Stateful connections change your architecture.

- Sticky sessions or a pub/sub backplane (Redis): a message for user X must reach the server holding X's socket.
- Connection counts: 100k idle sockets is normal — tune file descriptors, use event loops (not thread-per-connection).
- Proxies must support Upgrade (most do now); set generous idle timeouts or send heartbeats.
- Auth once at handshake (token in query/header) — then trust the socket, but revalidate on sensitive actions.

wss:// (TLS) everywhere in production — ws:// leaks everything and mixed-content rules will block it anyway.

STEP ✓/✓

Key Takeaways

WebSockets — Remember

- HTTP Upgrade handshake → 101 Switching Protocols → persistent framed connection.
- Frames carry opcodes (text/binary/ping/pong); client frames are masked.
- Full-duplex: either side sends anytime — ideal for chat, live data, games.
- SSE for server→client feeds; long-polling for max compatibility; WebSocket for bidirectional.
- Scale with a pub/sub backplane, heartbeats, and wss:// always.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev